Discrete Reward-tilting

Jaemoo Choi

- **Lecture 5** The four methods of Lectures 3 and 4, rewritten for a continuous-time Markov chain on a finite state space: Feynman–Kac correctors (no training), weighted target concrete score matching and its proximal variant (train the rate from weighted endpoints), the discrete adjoint Schrödinger bridge sampler and the discrete bridge matching sampler (solve for the rate when the target is an energy), and the extension to diffusion language models.

Date: August 31, 2026

1 The Discrete Reward-Tilting Problem

Notations $\mathcal{X} = [N]^D$: the state space — sequences of D coordinates with N values each, finite throughout; x^d is the d-th coordinate of x and $x^{d\leftarrow n}$ is x with that coordinate set to n. μ: the law at $t = 0$ (the fully masked sequence, or the uniform law on $\mathcal{X}$). p^{base}: the reference path measure on $[0, 1]$, with marginals p_t^{base}, transitions $p_{t|s}^{\text{base}}(y \mid x)$, denoising posterior $p_{1|t}^{\text{base}}(x_1 \mid x)$ and noising kernel $p_{t|1}^{\text{base}}(x \mid x_1)$. $r : \mathcal{X} \to \mathbb{R}$: the reward — by *value* only, since a finite state space has no gradient; $\beta > 0$: the inverse temperature. F: a test function. Time runs from noise ($t = 0$) to data ($t = 1$). Two unrelated objects share the letter γ, each carrying its own index and never appearing together: γ_k is the proximal parameter of theorem 3.6, while γ_t is the noise schedule of the uniform reference (32), introduced in Section 4 and used again in Section 6.

Rates replace drifts The one structural change from Lecture 3 is that the chain is specified by transition *rates* rather than by a drift, and a rate carries *two* arguments.

Definition 1.1 (The chain and its rates)**.** A rate $b_t(y, x) \geq 0$ ($y \neq x$) is the intensity of a jump *from x to y* at time t:

$$\Pr\left(X_{t+h} = y \,\middle|\, X_t = x\right) = \delta(y, x) + b_t(y, x)\, h + o(h), \qquad b_t(x, x) := -\sum_{y\neq x} b_t(y, x), \tag{1}$$

δ being the Kronecker delta, so that the rows of the rate matrix sum to zero. Its generator is $\mathcal{B}_t F(x) = \sum_{y\neq x} b_t(y, x)[F(y) - F(x)]$, and its marginals solve the master equation

$$\partial_t\, p_t^{\text{base}}(x) = \sum_{y\neq x} \Big[b_t(x, y)\, p_t^{\text{base}}(y) - b_t(y, x)\, p_t^{\text{base}}(x)\Big]. \tag{2}$$

Setting: the pretrained model Given is a discrete diffusion model, that is, its generative rates $b_t(y, x)$; p^{base} denotes the law of the chain they generate from $X_0 \sim \mu$. In Sections 2 and 3 the reference *is* that pretrained model; in Sections 4 and 5 it is a uniform chain and no pretrained model exists at all. One reference letter serves both roles.

Definition 1.2 (The reward-tilted target)**.**

$$p_1^\star(x) \;\propto\; p_1^{\text{base}}(x)\, e^{\beta r(x)}; \qquad \text{on path space} \qquad \frac{\mathrm{d}p^\star}{\mathrm{d}p^{\text{base}}}(X) = \frac{e^{\beta r(X_1)}}{Z}, \qquad Z = \mathbb{E}_{p^{\text{base}}}\left[e^{\beta r(X_1)}\right]. \tag{3}$$

Definition 1.3 (The proposal)**.** For rates $u_t(y, x) \geq 0$ with $u_t = 0$ wherever $b_t = 0$, the proposal p^u is the law of the chain with those rates started at $X_0 \sim \mu$. The control *is* the rate. Its generator is written $\mathcal{U}_t$.

Remark (The two levers)**.** The network is frozen: as in Lecture 3, what remains is the chain we simulate and a weight we carry. Everything below is one identity for that weight, used in three ways — carried (§2), used as an importance weight for training (§3), or minimized in u (§4–§5).

1.1 Dynkin's Formula

Theorem 1.4 (Itô's formula for a chain, and the compensator)**.** *Along a trajectory of p^u, for a function $F(t,x)$ that is C^1 in t,*

$$F(1,X_1) - F(0,X_0) = \int_0^1 \big(\partial_t + \mathcal{U}_t\big)F(t,X_t)\,\mathrm{d}t \;+\; M_1, \tag{4}$$

and, for any $\Psi_t(y,x)$,

$$\mathbb{E}\sum_{\textit{jumps } t} \Psi_t\big(X_t, X_{t-}\big) = \mathbb{E}\int_0^1 \sum_{y\neq X_t} u_t(y,X_t)\,\Psi_t(y,X_t)\,\mathrm{d}t, \tag{5}$$

M being in both cases the jump sum minus its compensator — a mean-zero martingale.

Proof. (i) A pathwise identity. $\mathcal{X}$ is finite, so a trajectory has finitely many jump times $0<\tau_1<\cdots<\tau_K<1$. On $[\tau_k,\tau_{k+1})$ the state is constant, so $\frac{\mathrm{d}}{\mathrm{d}t}F(t,X_t) = \partial_t F(t,X_t)$; integrating on each piece and telescoping,

$$F(1,X_1) - F(0,X_0) = \int_0^1 \partial_t F(t,X_t)\,\mathrm{d}t + \sum_{\text{jumps } t}\Big[F(t,X_t) - F\big(t,X_{t-}\big)\Big]. \tag{6}$$

Nothing probabilistic has been used: this holds trajectory by trajectory.

(ii) The jump sum is compensated. Put $M_t := \sum_{\text{jumps}\le t}\Psi - \int_0^t \sum_{y\neq X_s} u_s(y,X_s)\Psi_s(y,X_s)\mathrm{d}s$. From (1), $\Pr(\text{a jump to } y \text{ in } (t,t+\Delta t] \mid X_t = x) = u_t(y,x)\Delta t + o(\Delta t)$ and $\Pr(\text{two or more}) = o(\Delta t)$, so the two contributions to $\mathbb{E}[M_{t+\Delta t} - M_t \mid X_t = x]$ cancel to $o(\Delta t)$: M is a martingale and $\mathbb{E}M_1 = 0$, which is (5). Taking $\Psi_t(y,x) = F(t,y) - F(t,x)$ in (6) turns the jump sum into $\int_0^1 \mathcal{U}_t F\,\mathrm{d}t + M_1$, which is (4). ☐

1.2 Girsanov's Theorem

Theorem 1.5 (Girsanov for Markov chains)**.** *Write $\lambda_t^u(x) = \sum_{y\neq x} u_t(y,x)$ for the total jump rate. Then*

$$\log\frac{\mathrm{d}p^u}{\mathrm{d}p^{\text{base}}}(X) = \sum_{\textit{jumps } t}\log\frac{u_t}{b_t}\big(X_t,X_{t-}\big) \;-\; \int_0^1 \sum_{y\neq X_t}\big(u_t - b_t\big)(y,X_t)\,\mathrm{d}t, \tag{7}$$

and, in expectation,

$$D_{\mathrm{KL}}\big(p^u \,\big\|\, p^{\text{base}}\big) = \mathbb{E}_{p^u}\int_0^1 \sum_{y\neq X_t}\Big[u_t\log\tfrac{u_t}{b_t} - u_t + b_t\Big](y,X_t)\,\mathrm{d}t. \tag{8}$$

Proof. (i) One step. Over $(t,t+\Delta t]$ the chain either waits or jumps once, up to $o(\Delta t)$. The no-jump probabilities are $1-\lambda_t^u\Delta t$ and $1-\lambda_t^b\Delta t$, whose ratio is $e^{-(\lambda_t^u - \lambda_t^b)\Delta t} + o(\Delta t)$; the jump probabilities are $u_t(y,x)\Delta t$ and $b_t(y,x)\Delta t$, whose ratio is $u_t(y,x)/b_t(y,x)$ — *the Δt cancels at a jump*, and only the ratio of the rates survives.

(ii) Multiply over a partition. The waiting factors exponentiate into $\exp(-\int_0^1(\lambda_t^u - \lambda_t^b)(X_t)\mathrm{d}t)$ and the jump factors multiply, giving $\frac{\mathrm{d}p^u}{\mathrm{d}p^{\text{base}}}(X) = \prod_{\text{jumps}}\frac{u_t}{b_t}\,\exp(-\int_0^1\sum_{y\neq X_t}(u_t - b_t)\mathrm{d}t)$; take logarithms for (7).

(iii) The divergence. Take $\mathbb{E}_{p^u}$ and compensate the jump sum by (5) with $\Psi_t = \log\frac{u_t}{b_t}$: the jump sum becomes $\mathbb{E}_{p^u}\int_0^1\sum_{y\neq X_t} u_t\log\frac{u_t}{b_t}\mathrm{d}t$, which is (8). A jump is paid for by a log-ratio and waiting by the total-rate gap. Each summand $u\log\frac{u}{b} - u + b$ is ≥ 0 and vanishes only at $u=b$, so (8) is genuinely a divergence. ☐

Definition 1.6 (The interpolated reward)**.** Choose any $r_t : \mathcal{X}\to\mathbb{R}$, smooth in t, with $r_0 \equiv 0$ and $r_1 = \beta r$ — the reward credited by time t, so that the reference tilted by e^{r_t} starts at the reference and ends at the target:

$$r_0 \equiv 0, \qquad r_1 = \beta r, \qquad \mathrm{d}p^\star \propto e^{r_1(X_1)}\,\mathrm{d}p^{\text{base}}. \tag{9}$$

The running example throughout is $r_t = \beta_t r$ with $\beta_0 = 0$, $\beta_1 = \beta$.

2 Discrete Feynman–Kac Correctors

2.1 The Tilting Identity

Theorem 2.1 (The tilting identity)**.** *Define the* tilted rate

$$U_t(y,x) := e^{\,r_t(y)-r_t(x)}\,b_t(y,x), \qquad y \neq x. \tag{10}$$

Then, for every *rate* u*,*

$$\begin{aligned}\mathrm{d}p^\star(X) \;&\propto\; \exp\Big(\int_0^1 L_t\big(X_t,u_t\big)\mathrm{d}t + M_1(X)\Big)\,\mathrm{d}p^u(X),\\ L_t(x,u) &= \partial_t r_t(x) + \sum_{y\neq x}\Big[e^{\,r_t(y)-r_t(x)} - 1\Big]b_t(y,x) - \mathcal{D}\big(u_t \,\|\, U_t\big)(x),\end{aligned} \tag{11}$$

where M is the compensated jump martingale of theorem 1.4 for $\Psi_t = \log\frac{U_t}{u_t}$, written out

$$M_1(X) = \sum_{\textit{jumps } t} \log\frac{U_t}{u_t}\big(X_t, X_{t-}\big) - \int_0^1 \sum_{y\neq X_t} u_t(y,X_t)\,\log\frac{U_t}{u_t}(y,X_t)\,\mathrm{d}t \tag{12}$$

— mean zero under p^u, and identically zero at $u = U$ — and

$$\mathcal{D}\big(u\,\|\,U\big)(x) := \sum_{y\neq x}\Big[u\log\tfrac{u}{U} - u + U\Big](y,x) \;\geq\; 0, \quad \textit{zero iff } u = U. \tag{13}$$

Proof. (i) Split the weight. By (9) and the chain rule, $\log\frac{\mathrm{d}p^\star}{\mathrm{d}p^u}(X) = r_1(X_1) + \log\frac{\mathrm{d}p^{\text{base}}}{\mathrm{d}p^u}(X) + \text{const}$, and theorem 1.5 with the sign reversed gives the second term.

(ii) Write $r_1(X_1)$ along the path. Since $r_0 \equiv 0$, the map $t \mapsto r_t(X_t)$ starts at 0, is smooth between jumps and steps at a jump, so by (6)

$$r_1(X_1) = \int_0^1 \partial_t r_t(X_t)\,\mathrm{d}t + \sum_{\text{jumps } t}\Big[r_t(X_t) - r_t\big(X_{t-}\big)\Big].$$

At a jump $x \to y$ the two jump sums combine into a single term, and this is what names U: $r_t(y) - r_t(x) + \log\frac{b_t(y,x)}{u_t(y,x)} = \log\frac{U_t(y,x)}{u_t(y,x)}$.

(iii) Collect and compensate. Adding the two displays,

$$\log\frac{\mathrm{d}p^\star}{\mathrm{d}p^u}(X) = \int_0^1 \Big[\partial_t r_t + \sum_{y\neq X_t}\big(u_t - b_t\big)\Big](X_t)\mathrm{d}t + \sum_{\text{jumps } t}\log\frac{U_t}{u_t} + \text{const}. \tag{14}$$

Compensate the last jump sum by (5) with $\Psi_t = \log\frac{U_t}{u_t}$; it contributes $\int_0^1\sum_{y\neq X_t} u_t \log\frac{U_t}{u_t}\mathrm{d}t + M_1$. Adding and subtracting $\sum_{y\neq x} U_t$ in the integrand separates it into the u-free part of (11) and $-\mathcal{D}(u_t\|U_t)$ of (13). $\square$

Remark (The dictionary)**.** One line of Lecture 3 changes — the tilt is multiplicative, not a gradient.

Lecture 3 (diffusion)	**Lecture 5 (Markov chain)**
control $u_t(x)$, drift $b_t + \sigma_t u_t$	rate $u_t(y, x)$
$U_t = \sigma_t \nabla r_t$	$U_t(y, x) = e^{\,r_t(y) - r_t(x)}\, b_t(y, x)$
$\frac{1}{2}\Vert u_t - U_t \Vert^2$	$\sum_{y \neq x} \big[u \log(u/U) - u + U\big]$
$\partial_t + b_t \cdot \nabla + \frac{\sigma_t^2}{2}\Delta$	$\partial_t + \mathcal{B}_t$
$G_t^\top \mathrm{d}W_t$	the compensated jump martingale $\mathrm{d}M_t$

The tilted rate and the residual U_t is a Doob h-transform of the reference by e^{r_t}: it steers up the tilt, and it is the exact discrete counterpart of Lecture 2's $u_t^\star = \sigma_t \nabla \log \varphi_t$ with e^{r_t} in place of φ_t. The u-free part of (11) is the martingale residual of the chosen interpolation,

$$\partial_t r_t(x) + \sum_{y \neq x} \Big[e^{\,r_t(y) - r_t(x)} - 1\Big] b_t(y, x) = \frac{(\partial_t + \mathcal{B}_t)e^{r_t}}{e^{r_t}}(x), \tag{15}$$

and it vanishes identically if and only if $e^{r_t(X_t)}$ is a p^{base}-martingale, that is, iff $e^{r_t} = \mathbb{E}_{p^{\text{base}}}\big[e^{\beta r(X_1)} \mid X_t\big]$ — the value function. Steering alone does not remove it: at $u = U$ the divergence and the jump martingale both die, and $\mathrm{d}p^\star \propto \exp\big(\int_0^1 \big[\partial_t r_t + \sum_{y \neq X_t}(U_t - b_t)(y, X_t)\big](X_t)\,\mathrm{d}t\big)\,\mathrm{d}p^U$ — the noise term is gone, the drift remains.

2.2 The Weight Process and the Sampler

Corollary 2.2 (The weight of the current rate)**.** *Simulate the pair* $(X_t, \log w_t)$ *along one trajectory of* p^u*, with* $w_0 = 1$*:*

$$\mathrm{d} \log w_t = \Big[\partial_t r_t + \sum_{y \neq X_t} \big(u_t - b_t\big)\Big](X_t)\,\mathrm{d}t \;+\; \log \frac{U_t}{u_t}\big(X_t, X_{t^-}\big)\Big|_{\text{at each jump}}. \tag{16}$$

Then $w_1 \propto \frac{\mathrm{d}p^\star}{\mathrm{d}p^u}(X)$ *and, for every test function* F*,*

$$\frac{\mathbb{E}_{p^u}\big[w_1 F(X_1)\big]}{\mathbb{E}_{p^u}\big[w_1\big]} = \mathbb{E}_{p_1^\star}\big[F\big]. \tag{17}$$

Proof. Integrating (16) reproduces (14) exactly, so $\log w_1 = \log \frac{\mathrm{d}p^\star}{\mathrm{d}p^u}(X) + \log Z$ for the path-free constant Z. Hence $\mathbb{E}_{p^u}[w_1 F(X_1)] = Z\,\mathbb{E}_{p^\star}[F(X_1)]$; taking $F \equiv 1$ gives $\mathbb{E}_{p^u}[w_1] = Z$, and the ratio cancels the unknown Z. □

Remark (Why the jumps are left uncompensated)**.** (16) is theorem 2.1 with the jump sum *not* compensated. That is the implementable form: every term is read off the simulated trajectory, and nothing — neither M nor $\mathcal{D}$ — has to be estimated.

Algorithm 1 Discrete Feynman–Kac correctors (Hasan et al., 2026; Wu et al., 2023)

1: Draw K particles $X_0^i \sim \mu$ and set $\log w_0^i = 0$.
2: τ-leap every particle — coordinates jump independently, $\Pr(X_{t+\Delta t} = x^{d \leftarrow n} \mid X_t = x) = u_t(x^{d \leftarrow n}, x)\Delta t$ — and update its log-weight by (16).
3: Resample the particles by $\mathrm{softmax}_i(\log w_t^i)$ and reset the weights; this is legal at every step.
4: At $t = 1$ report $\sum_i w_1^i F(X_1^i) \big/ \sum_i w_1^i$.

The reward-tilted chain For the running example $r_t = \beta_t r$ everything in (11) is explicit:

$$\begin{aligned} U_t(y, x) &= e^{\beta_t [r(y) - r(x)]}\, b_t(y, x), \\ L_t(x, u) &= \dot{\beta}_t\, r(x) + \sum_{y \neq x} \Big[e^{\beta_t [r(y) - r(x)]} - 1\Big] b_t(y, x) - \mathcal{D}\big(u_t \,\Vert\, U_t\big)(x). \end{aligned} \tag{18}$$

Both lines need only the model's rates and $N-1$ reward *differences* per coordinate. Annealing, products of experts and reward tilts are the same formula with another r_t.

Three interpolated rewards The freedom in definition 1.6 is the freedom in r_t, and the three standard choices are

$$r_t = \beta_t\, r; \qquad r_t(x) = \beta_t\, \mathbb{E}_{p^{\text{base}}}\big[r(X_1) \mid X_t = x\big]; \qquad r_t(x) = \log \mathbb{E}_{p^{\text{base}}}\big[e^{\beta r(X_1)} \mid X_t = x\big]. \tag{19}$$

A masked or heavily noised state has no reward of its own, so the last two ask the denoiser instead; the third is the value function, for which the residual (15) vanishes and the weight is constant at $u = U$. Tractability is never the issue — (15) is explicit for every explicit r_t — weight variance is.

Remark (What the correctors buy and what they cost)**.** *Advantage:* no training. The network stays frozen, any reward and any β are available at inference time, and resampling is legal at every step. *Limitation:* the tilted rate costs $D(N-1)$ reward evaluations per step, and a poor proposal is paid in weight variance — the effective sample size $K_{\text{eff}} = 1/\sum_i (w^i)^2$ collapses. The remedy is to train the rate, which is the next two sections.

3 Cross Entropy: Weighted Target Concrete Score Matching

3.1 The Cross-Entropy Problem

Definition 3.1 (The trainee and the objective)**.** The network predicts the tilt *ratio*, not the rate:

$$p^{u^\theta}: \quad u_t^\theta(y,x) = \Phi_t^\theta(x)_y\, b_t(y,x), \quad X_0 \sim \mu; \qquad \min_\theta\ D_{\text{KL}}\big(p^\star \,\big\|\, p^{u^\theta}\big) = \min_\theta\ \mathbb{E}_{X\sim p^\star}\big[-\log p^{u^\theta}(X)\big]. \tag{20}$$

This is the *forward* KL — the unknown first — so the expectation is over the target, which we cannot sample.

Proposition 3.2 (The importance-weighted cross entropy)**.** *Let $\bar{u}$ be any rate. Then*

$$\mathbb{E}_{X\sim p^\star}\big[-\log p^{u^\theta}(X)\big] = \mathbb{E}_{X\sim p^{\bar{u}}}\Big[-\frac{\mathrm{d}p^\star}{\mathrm{d}p^{\bar{u}}}(X)\,\log p^{u^\theta}(X)\Big], \tag{21}$$

$$\log \frac{\mathrm{d}p^\star}{\mathrm{d}p^{\bar{u}}}(X) = \beta r(X_1) - \sum_{\textit{jumps } t} \log \frac{\bar{u}_t}{b_t}(X_t, X_{t^-}) + \int_0^1 \sum_{y \neq X_t} \big(\bar{u}_t - b_t\big)(y, X_t)\,\mathrm{d}t - \log Z. \tag{22}$$

Proof. (21) is a change of measure. For (22), split through the reference: $\frac{\mathrm{d}p^\star}{\mathrm{d}p^{\bar{u}}} = \frac{\mathrm{d}p^\star}{\mathrm{d}p^{\text{base}}} \cdot \frac{\mathrm{d}p^{\text{base}}}{\mathrm{d}p^{\bar{u}}}$, the first factor being $e^{\beta r(X_1)}/Z$ by (3) and the second (7) with the sign reversed. □

Remark (The default choice)**.** With $\bar{u} = b$ — simulate the pretrained model itself — every path term drops and the weight is just $\beta r(X_1) - \log Z$, which the softmax normalizes away.

3.2 The Target Concrete Score

Why a ratio is the right object A law on a finite space is determined by its ratios at neighbouring states, the *concrete score*. On clean data the tilt makes that ratio elementary,

$$\frac{p_1^\star\big(x_1^{d\leftarrow n}\big)}{p_1^\star(x_1)} = \frac{p_1^{\text{base}}\big(x_1^{d\leftarrow n}\big)}{p_1^{\text{base}}(x_1)}\, e^{\beta[r(x_1^{d\leftarrow n}) - r(x_1)]}, \tag{23}$$

with Z cancelling and the reward evaluated on clean x_1 only, never on a noised state. What has to be learned is the same ratio at time t.

Theorem 3.3 (The concrete score of the target)**.** *Let* $\varphi_t(x) := \mathbb{E}_{p^{\text{base}}}[e^{\beta r(X_1)} \mid X_t = x]$*, so that* $p_t^\star \propto p_t^{\text{base}}\varphi_t$ *and the optimal rate is* $u_t^\star(y,x) = (\varphi_t(y)/\varphi_t(x))b_t(y,x)$*. Then, for all* $y \neq x$*,*

$$\frac{\varphi_t(y)}{\varphi_t(x)} = \mathbb{E}_{p^\star_{1|t}(x_1|x)}\left[\frac{p^{\text{base}}_{1|t}(x_1 \mid y)}{p^{\text{base}}_{1|t}(x_1 \mid x)}\right]. \tag{24}$$

Proof. Write the definition out, $\varphi_t(x) = \sum_{x_1} p^{\text{base}}_{1|t}(x_1 \mid x)e^{\beta r(x_1)}$. Because $p^\star$ is an endpoint tilt of p^{base}, Bayes gives $p^\star_{1|t}(x_1 \mid x) = p^{\text{base}}_{1|t}(x_1 \mid x)e^{\beta r(x_1)}/\varphi_t(x)$. Substituting into the right-hand side of (24) cancels $p^{\text{base}}_{1|t}(x_1 \mid x)$ and leaves $\frac{1}{\varphi_t(x)}\sum_{x_1} p^{\text{base}}_{1|t}(x_1 \mid y)e^{\beta r(x_1)} = \varphi_t(y)/\varphi_t(x)$. □

Remark (The discrete stand-in)**.** (24) plays the role of Lecture 3's $\nabla \log p_t^\star(x) = \mathbb{E}[\nabla \log p^{\text{base}}_{t|1} \mid X_t = x]$: a conditional expectation of a *reference* kernel ratio under the target's own endpoint law. It is a training identity, not a sampler — the target is $p_1^\star \propto p_1^{\text{base}} e^{\beta r}$ and only reward evaluations are needed (Zhang et al., 2025).

Theorem 3.4 (The weighted denoising objective)**.** *Draw* $X^1, \dots, X^K \sim p^{\bar{u}}$ *and set* $w^i = \operatorname{softmax}_i\big(\log \frac{\mathrm{d}p^\star}{\mathrm{d}p^{\bar{u}}}(X^i)\big)$ *by* (22)*. Then the minimizer of*

$$\mathcal{L}(\theta) = \sum_{i=1}^{K} w^i \, \mathbb{E}_{t,\,X_t} \sum_{d=1}^{D} \sum_{n \neq X_t^d} \mathcal{D}\left(\frac{p^{\text{base}}_{1|t}(X_1^i \mid X_t^{d\leftarrow n})}{p^{\text{base}}_{1|t}(X_1^i \mid X_t)} \,\Big\|\, \Phi_t^\theta(X_t)_{d,n}\right), \tag{25}$$

with $t \sim \operatorname{Unif}[0,1]$ *and* $X_t \sim p^{\text{base}}_{t|1}(\cdot \mid X_1^i)$*, is* $\Phi_t(x)_{d,n} = \varphi_t(x^{d\leftarrow n})/\varphi_t(x)$*, that is* $u^\theta = u^\star$*.*

Proof. By theorem 1.5 applied to the pair, $D_{\mathrm{KL}}(p^\star \| p^{u^\theta}) = \mathbb{E}_{p^\star}\int_0^1 \mathcal{D}(u_t^\star \| u_t^\theta)(X_t)\mathrm{d}t$, so the training problem is a $\mathcal{D}$-regression along $p^\star$. Because $p^\star$ is an endpoint tilt, the bridges agree, $p^\star_{t|1} = p^{\text{base}}_{t|1}$, so sampling $(X_t, X_1) \sim p^\star_{t,1}$ is the same as drawing $X_1 \sim p_1^\star$ and noising it back, $X_t \sim p^{\text{base}}_{t|1}(\cdot \mid X_1)$ — which is exactly the sampling in (25), the weights w^i standing for $p_1^\star$ by (21). Finally $\mathcal{D}$ is a Bregman divergence, hence minimized at the conditional mean of its target: the minimizer is $\mathbb{E}[p^{\text{base}}_{1|t}(X_1 \mid X_t^{d\leftarrow n})/p^{\text{base}}_{1|t}(X_1 \mid X_t) \mid X_t]$, which is $\varphi_t(X_t^{d\leftarrow n})/\varphi_t(X_t)$ by theorem 3.3. □

Remark (Masked diffusion)**.** For a masked reference the kernel ratio degenerates and (25) is the familiar weighted denoising cross-entropy on the masked coordinates.

Algorithm 2 Weighted target concrete score matching (Zhang et al., 2025)

1: τ-leap K particles with the rates $\bar{u}_t$ from $X_0^i \sim \mu$, accumulating the Girsanov log-weight of (22): subtract $\log \frac{\bar{u}_t}{b_t}(X_t^i, X_{t-}^i)$ at a jump, add $\sum_{y \neq X_t^i}(\bar{u}_t - b_t)\Delta t$ while waiting.
2: At $t = 1$ set $w^i = \operatorname{softmax}_i\big(\log w^i + \beta r(X_1^i)\big)$; $\log Z$ drops out. Keep the pairs (X_1^i, w^i).
3: Draw $t \sim \operatorname{Unif}[0,1]$ and $X_t \sim p^{\text{base}}_{t|1}(\cdot \mid X_1^i)$ — the endpoint noised back to time t.
4: Take a gradient step on (25); optionally set $\bar{u} \leftarrow u^\theta$ and return to step 1.

3.3 Two Extensions

Proposition 3.5 (Extension I: weights from any corrector sampler)**.** *Any pairs* (X_1^i, w^i) *produced by algorithm 1 — any interpolation* r_t*, any rate* u*, resampling included — form a valid weighted batch for* (25)*.*

Proof. $\mathcal{L}$ consumes only the endpoints: it needs (X_1, w) with $\mathbb{E}[wF(X_1)]/\mathbb{E}[w] = \mathbb{E}_{p_1^\star}[F]$ for every F, which is exactly (17). The path measures differ, the endpoint marginal does not; the sampler's choices move variance, never correctness. □

Theorem 3.6 (Extension II: the proximal cross entropy)**.** *Let $p^{(k)}$ be the current model (rates $u^{(k)}$, $p^{(0)} = p^{\text{base}}$, $u^{(0)} = b$) and $\gamma_k > 0$. Then*

$$p^{(k+1)} := \arg\min_p \Big\{ D_{\text{KL}}\big(p \,\|\, p^\star\big) + \tfrac{1}{\gamma_k} D_{\text{KL}}\big(p \,\|\, p^{(k)}\big) \Big\} \quad \Longrightarrow \quad \mathrm{d}p^{(k+1)} \propto \big(\mathrm{d}p^\star\big)^{\alpha_k} \big(\mathrm{d}p^{(k)}\big)^{1-\alpha_k}, \tag{26}$$

with $\alpha_k = \gamma_k/(1+\gamma_k)$, and the weight for the inner target, sampling $X \sim p^{(k)}$, is

$$\log w = \alpha_k \Big(\beta r(X_1) - \log \frac{\mathrm{d}p^{(k)}}{\mathrm{d}p^{\text{base}}}(X) \Big) + \text{const.} \tag{27}$$

Proof. Write $\alpha = \alpha_k$, so that α and $1-\alpha$ sum to one. Then

$$D_{\text{KL}}\big(p \,\|\, p^\star\big) + \tfrac{1}{\gamma_k} D_{\text{KL}}\big(p \,\|\, p^{(k)}\big) = \Big(1 + \tfrac{1}{\gamma_k}\Big) \mathbb{E}_p \log \frac{\mathrm{d}p}{\big(\mathrm{d}p^\star\big)^{\alpha} \big(\mathrm{d}p^{(k)}\big)^{1-\alpha}} = \Big(1 + \tfrac{1}{\gamma_k}\Big) \Big[D_{\text{KL}}\big(p \,\|\, p^{(k+1)}\big) - \log Z_k \Big],$$

Z_k being the normalizer of (26): the two divergences merge into one, and $D_{\text{KL}} \geq 0$ finishes it. For the weight, $\frac{\mathrm{d}p^{(k+1)}}{\mathrm{d}p^{(k)}} \propto (\frac{\mathrm{d}p^\star}{\mathrm{d}p^{(k)}})^\alpha$ by (26), and the bracket is $\log \frac{\mathrm{d}p^\star}{\mathrm{d}p^{(k)}}$ up to a constant by proposition 3.2 applied to $\bar{u} = u^{(k)}$. ☐

Algorithm 3 Proximal cross-entropy training on a chain (Guo et al., 2026b)

1. **outer loop** ($k = 0, 1, 2, \ldots$; $u^{(0)} = b$): τ-leap $X^1, \ldots, X^K \sim p^{(k)}$ with the rates $u^{(k)}$, accumulating $\log \frac{\mathrm{d}p^{(k)}}{\mathrm{d}p^{\text{base}}}$ by (7).
2. Set $w^i = \operatorname{softmax}_i\big(\alpha_k [\, \beta r(X_1^i) - \log \frac{\mathrm{d}p^{(k)}}{\mathrm{d}p^{\text{base}}}(X^i) \,]\big)$ and keep the pairs (X_1^i, w^i) — a batch for the inner target $p^{(k+1)}$.
3. **inner loop**: with the data fixed, draw $t \sim \text{Unif}[0,1]$ and $X_t \sim p^{\text{base}}_{t|1}(\cdot \mid X_1^i)$ and take gradient steps on (25) until converged.
4. Set $u^{(k+1)} \leftarrow u^\theta$ and return to step 1.

Remark (Why the proximal step)**.** One round of algorithm 2 asks the weights to bridge $p^{\bar{u}}$ all the way to $p^\star$, and the softmax then concentrates on a few sequences. The proximal step tempers the same weight by α_k and moves the model a little at a time, keeping K_{eff} healthy; the cost is many outer rounds.

3.4 The Soft Cross-Entropy

An exact local label instead of a path weight Everything so far weights whole sequences. When the target is an energy, $\nu \propto e^{-E}$, and the reference is a *masked* chain with reveal schedule κ_t and mask symbol $\mathtt{M}$, one can instead regress against the target's own single-coordinate conditional — computable exactly, with no weight at all (Blessing et al., 2026a). Write $C_t^\theta(x)_{d,n} \approx \Pr(X_1^d = n \mid X_t = x)$ for the model's denoising row, which fixes the rate by

$$u_t^\theta\big(x^{d \leftarrow n}, x\big) = \frac{\dot{\kappa}_t}{1 - \kappa_t}\, C_t^\theta(x)_{d,n}. \tag{28}$$

Theorem 3.7 (The minimizer of the soft cross-entropy)**.** *Let X_1 be drawn from the current model $p_1^{u^\theta}$ and let each coordinate be revealed with probability κ_t, and let*

$$\mathcal{L}(\theta) = -\,\mathbb{E} \sum_{d \,:\, X_t^d = \mathtt{M}} \sum_{n=1}^{N} \frac{\nu\big(X_1^{d \leftarrow n}\big)}{\sum_m \nu\big(X_1^{d \leftarrow m}\big)} \log C_t^\theta(X_t)_{d,n}. \tag{29}$$

At a state x with $x^d = \mathtt{M}$ its minimizer is $C_t^\theta(x)_{d,n} = \mathbb{E}\big[\nu(X_1^{d \leftarrow n}) / \sum_m \nu(X_1^{d \leftarrow m}) \mid X_t = x\big]$, and if $p_1^{u^\theta} = \nu$ this equals $\nu(X_1^d = n \mid X_t = x)$: the target is a fixed point of generate $\to$ label $\to$ regress.

Proof. *(i) The minimizer.* At the state x the objective is $-\sum_n \mathbb{E}[\,\cdot \mid X_t = x] \log C_t^\theta(x)_{d,n}$ over a probability row, whose minimizer is the row of conditional means (Gibbs' inequality).

(ii) The label is a conditional of ν. $\nu(x_1^{d\leftarrow n})/\sum_m \nu(x_1^{d\leftarrow m}) = \nu(X_1^d = n \mid x_1^j,\ j \neq d)$ — one coordinate given the others, and Z cancels.

(iii) Tower. If $p_1^{u^\theta} = \nu$, then given $X_t = x$ the endpoint is ν-distributed on the revealed coordinates, and masking ignores the symbols; the label of (ii) never reads X_1^d, so the tower property gives $\mathbb{E}[\nu(X_1^d = n \mid X_1^j, j \neq d) \mid X_t = x] = \nu(X_1^d = n \mid X_t = x)$. ☐

Algorithm 4 The soft cross-entropy (Blessing et al., 2026a)

1: *Generate*: τ-leap the current model from the all-mask state and keep the endpoints $\mathrm{sg}[X_1]$ — no sample of ν is used anywhere.
2: *Label*: one batched call to E gives the N substituted energies $E(X_1^{d\leftarrow n})$ at every coordinate; store them with X_1.
3: *Interpolate*: draw $t \sim \mathrm{Unif}[0,1]$ and re-mask each coordinate of a stored X_1 with probability $1 - \kappa_t$.
4: *Regress*: take a gradient step on (29) at the masked coordinates and return to step 1.

The weighted soft cross-entropy theorem 3.7 identifies the minimizer with ν's posterior only *at* the fixed point; away from it the endpoints come from p^{u^θ} while the label comes from ν. Reweighting the batch repairs the mismatch for every sampler:

$$w^i = \mathrm{softmax}_i\Big[-E\big(X_1^i\big) - \log p_1^{u^\theta}\big(X_1^i\big)\Big], \qquad \mathcal{L}(\theta) = -\sum_{i=1}^{K} w^i \sum_{d\,:\,X_t^d = \mathtt{M}} \sum_{n=1}^{N} \frac{\nu\big(X_1^{i,\,d\leftarrow n}\big)}{\sum_m \nu\big(X_1^{i,\,d\leftarrow m}\big)} \log C_t^\theta(X_t)_{d,n}, \tag{30}$$

the log-likelihood $\log p_1^{u^\theta}$ being accumulated along the trajectory and Z cancelling in the softmax.

Remark (What the cross-entropy route buys and what it costs)**.** *Advantage:* only the endpoints matter — any sampler, any weights, one clean-data regression — and the reward enters by value, so no gradient is ever needed. *Limitation:* the weights live on whole sequences, so if $p^{\bar{u}}$ covers $p_1^\star$ poorly the softmax concentrates and K_{eff} collapses; the proximal rounds fix that at the price of many outer rounds. The soft cross-entropy escapes the weights altogether by using an exact *local* label — and when the target is an energy the weights can be removed entirely, which is the subject of the next two sections.

4 The Discrete Adjoint Schrödinger Bridge Sampler

4.1 The Problem and the Reference

Definition 4.1 (Sampling from an energy on a chain)**.** Given an energy E on $\mathcal{X}$, evaluable pointwise — and no data, no pretrained model —

$$\nu(x) = \frac{e^{-E(x)}}{Z}, \qquad Z = \sum_{x\in\mathcal{X}} e^{-E(x)} \ \text{ unknown;} \qquad \text{wanted: a chain with } p_1^u = \nu,\ X_0 \sim \mu = \mathrm{Unif}(\mathcal{X}). \tag{31}$$

The uniform reference The reference is now a uniform chain — the discrete analogue of adding noise. On $\mathcal{X} = \mathbb{Z}_N^D$, with d_H the Hamming distance and γ_t a noise schedule,

$$b_t(y,x) = \begin{cases} \gamma_t/N, & d_H(y,x) = 1, \\ -\gamma_t D(1-1/N), & y = x, \\ 0, & \text{otherwise:} \end{cases} \tag{32}$$

one coordinate at a time, every value equally likely. The consequence that the whole section uses is *additivity*: the kernel depends on the difference alone,

$$p^{\text{base}}_{1|t}(y \mid x) = q_t\big(y - x\big) \qquad \text{for some } q_t, \tag{33}$$

which is the group structure of $\mathbb{Z}_N^D$.

4.2 The Optimality System

Definition 4.2 (The discrete Schrödinger bridge)**.**

$$\min_u \; D_{\mathrm{KL}}\big(p^u \,\big\|\, p^{\text{base}}\big) \qquad \text{s.t.} \qquad p^u_0 = \mu, \quad p^u_1 = \nu, \tag{34}$$

with the objective given by (8). This is Lecture 2's problem with rates for drifts and $\mathcal{D}$ for $\frac{1}{2}\|u\|^2$.

Theorem 4.3 (The optimality system)**.** *The bridge is the h-transform of the reference by a pair of potentials* $(\varphi_t, \widehat{\varphi}_t)$*:*

$$u^\star_t(y, x) = \frac{\varphi_t(y)}{\varphi_t(x)}\, b_t(y, x), \qquad y \neq x, \tag{35}$$

$$\varphi_s(x) = \sum_y p^{\text{base}}_{t|s}(y \mid x)\, \varphi_t(y), \qquad \widehat{\varphi}_t(x) = \sum_y p^{\text{base}}_{t|s}(x \mid y)\, \widehat{\varphi}_s(y) \qquad \textit{(pull, push)}, \tag{36}$$

$$p^\star_t = \varphi_t \widehat{\varphi}_t, \qquad \varphi_0 \widehat{\varphi}_0 = \mu, \qquad \varphi_1 \widehat{\varphi}_1 = \nu. \tag{37}$$

The unknowns are therefore the endpoint pair $(\varphi_1, \widehat{\varphi}_1)$ *— exactly as in Lecture 2.*

Proof. (i) Feasibility. Repeat the proof of theorem 2.1 with φ_t in place of e^{r_t} — only the boundary changes, $\varphi_0 \not\equiv 1$ contributing $-\log \varphi_0(X_0)$. At $u = u^\star$ the divergence and the jump martingale die, and

$$\log \frac{\mathrm{d}p^{u^\star}}{\mathrm{d}p^{\text{base}}}(X) = \log \frac{\varphi_1(X_1)}{\varphi_0(X_0)} - \int_0^1 \frac{(\partial_t + \mathcal{B}_t)\varphi_t}{\varphi_t}(X_t)\, \mathrm{d}t = \log \frac{\varphi_1(X_1)}{\varphi_0(X_0)}, \tag{38}$$

the integrand vanishing because the pull equation in (36) is the backward equation $\partial_t \varphi_t + \mathcal{B}_t \varphi_t = 0$. Writing the time-$t$ marginal of that measure out and splitting the path at t,

$$p^\star_t(x) = \sum_{x_0, x_1} \frac{\mu(x_0)}{\varphi_0(x_0)}\, p^{\text{base}}_{t|0}(x \mid x_0)\, p^{\text{base}}_{1|t}(x_1 \mid x)\, \varphi_1(x_1) = \widehat{\varphi}_t(x)\, \varphi_t(x),$$

by the push equation on the left factor (with $\mu/\varphi_0 = \widehat{\varphi}_0$) and the pull equation on the right. At $t = 0$ and $t = 1$ this is (37), so the h-transform is feasible for (34).

(ii) Optimality. Let p be any path measure with $p_0 = \mu$ and $p_1 = \nu$. By (38),

$$D_{\mathrm{KL}}\big(p \,\|\, p^{\text{base}}\big) = D_{\mathrm{KL}}\big(p \,\|\, p^\star\big) + \mathbb{E}_\nu\big[\log \varphi_1\big] - \mathbb{E}_\mu\big[\log \varphi_0\big],$$

and the two expectations depend only on the fixed marginals, not on p. Evaluating the same identity at $p = p^\star$ identifies that constant as $D_{\mathrm{KL}}(p^\star \| p^{\text{base}})$, so

$$D_{\mathrm{KL}}\big(p \,\|\, p^{\text{base}}\big) = D_{\mathrm{KL}}\big(p \,\|\, p^\star\big) + D_{\mathrm{KL}}\big(p^\star \,\|\, p^{\text{base}}\big) \tag{39}$$

— a Pythagorean identity. The second term is fixed, so the minimum is at $p = p^\star$ and only there. □

4.3 The Control Formulation

Theorem 4.4 (One stage is one control problem)**.** *Fix the current backward potential $\widehat{\varphi}_1$ and set the terminal cost $g := \log(\widehat{\varphi}_1/\nu)$. Then the bridge rate solves the* unconstrained *problem*

$$\min_u \; D_{\mathrm{KL}}\big(p^u \,\big\|\, p^{\mathrm{base}}\big) + \mathbb{E}_{X\sim p^u}\big[g(X_1)\big] \qquad s.t. \quad p_0^u = \mu, \tag{40}$$

whose solution is the h-transform with $\varphi_1 = e^{-g} = \nu/\widehat{\varphi}_1$.

Proof. Put $\varphi_1 := e^{-g}$ and $\varphi_t(x) := \mathbb{E}_{p^{\mathrm{base}}}[\varphi_1(X_1) \mid X_t = x]$, the pull. Then $\frac{\mathrm{d}p^\star}{\mathrm{d}p^{\mathrm{base}}} = \varphi_1(X_1)/\varphi_0(X_0)$ by (38), so for any p started at μ,

$$D_{\mathrm{KL}}\big(p \,\|\, p^{\mathrm{base}}\big) + \mathbb{E}_p\big[g(X_1)\big] = \mathbb{E}_p\Big[\log \frac{\mathrm{d}p}{\mathrm{d}p^{\mathrm{base}}} - \log \varphi_1(X_1)\Big] = D_{\mathrm{KL}}\big(p \,\|\, p^\star\big) - \mathbb{E}_\mu\big[\log \varphi_0\big].$$

The last term is the same for all candidates, since they all start at μ; so the minimum is at $p = p^\star$ (Guo et al., 2026a). □

The two unknowns Since $\varphi_1 = \nu/\widehat{\varphi}_1$ and the energy is known, only $\widehat{\varphi}_1$ is missing, and the ratio the sampler needs is computable without Z:

$$\frac{\varphi_1(z)}{\varphi_1(x_1)} = e^{\,E(x_1)-E(z)}\;\frac{\widehat{\varphi}_1(x_1)}{\widehat{\varphi}_1(z)}. \tag{41}$$

Two networks carry the two potentials as *rows* indexed by (coordinate, value),

$$\Phi_t(x)_{d,n} \approx \frac{\varphi_t\big(x^{d\leftarrow n}\big)}{\varphi_t(x)}, \qquad \widehat{\Phi}(x)_{d,n} \approx \frac{\widehat{\varphi}_1\big(x^{d\leftarrow n}\big)}{\widehat{\varphi}_1(x)}, \qquad u_t\big(x^{d\leftarrow n}, x\big) = \frac{\gamma_t}{N}\,\Phi_t(x)_{d,n}. \tag{42}$$

4.4 The Two Matching Identities

Theorem 4.5 (The discrete adjoint)**.** *Under the additivity (33), for every t and every $y \neq x$,*

$$\frac{\varphi_t(y)}{\varphi_t(x)} = \mathbb{E}_{p^\star_{1|t}(x_1|x)}\left[\frac{\varphi_1\big(x_1 + y - x\big)}{\varphi_1(x_1)}\right]. \tag{43}$$

Proof. By the pull equation and (33), reindexing $x_1 = z + y - x$,

$$\varphi_t(y) = \sum_{x_1} q_t\big(x_1 - y\big)\varphi_1(x_1) = \sum_z q_t\big(z - x\big)\,\varphi_1\big(z + y - x\big) = \sum_z p^{\mathrm{base}}_{1|t}(z \mid x)\,\varphi_1\big(z + y - x\big).$$

The endpoint law of the h-transform is $p^\star_{1|t}(z \mid x) = p^{\mathrm{base}}_{1|t}(z \mid x)\varphi_1(z)/\varphi_t(x)$, so dividing by $\varphi_t(x)$ turns the sum into the stated expectation. The shift is legal only on a group: this is where $\mathbb{Z}_N^D$ is used. □

Remark (What replaced the gradient)**.** In Lecture 3 the adjoint state was $a_t = (\partial X_1/\partial X_t)^\top \beta \nabla r(X_1)$, obtained by integrating a backward ODE. Here the same first-order information is carried by a *shift of the endpoint* — no gradient, no backward pass, only differences of the energy.

Theorem 4.6 (The corrector)**.** *For every t and every y, z,*

$$\frac{\widehat{\varphi}_1(z)}{\widehat{\varphi}_1(y)} = \mathbb{E}_{p^\star_{t|1}(x|y)}\left[\frac{p^{\mathrm{base}}_{1|t}(z \mid x)}{p^{\mathrm{base}}_{1|t}(y \mid x)}\right] \qquad \textit{(bridge matching at } t = 1\textit{)}. \tag{44}$$

Proof. By Bayes and $p^\star_t = \varphi_t\widehat{\varphi}_t$, $p^\star_{t|1}(x \mid y) = p^\star_{1|t}(y \mid x)p^\star_t(x)/p^\star_1(y) = p^{\rm base}_{1|t}(y \mid x)\,\widehat{\varphi}_t(x)/\widehat{\varphi}_1(y)$ — every φ cancels. Hence

$$\sum_x p^\star_{t|1}(x \mid y)\,\frac{p^{\rm base}_{1|t}(z \mid x)}{p^{\rm base}_{1|t}(y \mid x)} = \frac{1}{\widehat{\varphi}_1(y)}\sum_x p^{\rm base}_{1|t}(z \mid x)\,\widehat{\varphi}_t(x) = \frac{\widehat{\varphi}_1(z)}{\widehat{\varphi}_1(y)},$$

the last step being the push equation of (36). □

The two losses Both identities are conditional means, so both are fitted by a $\mathcal{D}$-regression with the label first:

$$\min_{\Phi}\ \mathbb{E}_t\,\mathbb{E}_{p^\star_{t,1}} \sum_{d,\,n\neq x^d} \mathcal{D}\left(\frac{\varphi_1\big(x_1^{d\leftarrow x_1^d+n-x^d}\big)}{\varphi_1(x_1)}\ \Big\|\ \Phi_t(x)_{d,n}\right) \qquad \text{(adjoint matching)}, \tag{45}$$

$$\min_{\widehat{\Phi}}\ \mathbb{E}_t\,\mathbb{E}_{p^\star_{t,1}} \sum_{d,\,n\neq x_1^d} \mathcal{D}\left(\frac{p^{\rm base}_{1|t}\big(x_1^{d\leftarrow n} \mid x\big)}{p^{\rm base}_{1|t}\big(x_1 \mid x\big)}\ \Big\|\ \widehat{\Phi}(x_1)_{d,n}\right) \qquad \text{(corrector matching)}, \tag{46}$$

the φ_1 ratios in (45) being read off the energy and the current $\widehat{\Phi}$ through (41).

Algorithm 5 The discrete adjoint Schrödinger bridge sampler (Guo et al., 2026a)

1: Initialize $\widehat{\Phi} \equiv 1$, so that $\varphi_1 = \nu$ and the first stage is the pure tilt.
2: τ-leap K trajectories with the frozen rates $\mathrm{sg}[(\gamma_t/N)\Phi]$ and keep the endpoints x_1.
3: Draw $t \sim \mathrm{Unif}[0,1]$ and bridge: $x \sim p^{\rm base}_{t|0,1}(\cdot \mid x_0, x_1)$.
4: Fit the controller Φ by (45), with the φ_1 ratios read off E and $\mathrm{sg}[\widehat{\Phi}]$.
5: Fit the corrector $\widehat{\Phi}$ by (46) on the same pairs (x, x_1); return to step 2. At the fixed point the two networks are the bridge's potentials.

Remark (What the bridge costs)**.** No reward gradient, no likelihood, no backward simulation — only ratios. The price is two networks and two alternating objectives, and the requirement that the reference carry a group structure. The next section removes the first price by giving up the bridge itself.

5 The Discrete Bridge Matching Sampler

What is learned One network, and all it does is scale the reference's moves:

$$u_t\big(x^{d\leftarrow n}, x\big) = b_t\big(x^{d\leftarrow n}, x\big)\,\Phi_t(x)_{d,n}, \qquad \Phi > 0. \tag{47}$$

The loop is: run the current chain to $t = 1$ and keep the endpoint; apply the same move to that endpoint and score it by how much the energy drops; regress Φ on those scores; repeat. The rest of this section says what the score is, and why the fitted rate samples ν.

5.1 The Reciprocal Measure and its Projection

Definition 5.1 (The reciprocal measure)**.** Glue the reference's bridges under the *independent* coupling of the two marginals:

$$p^\star = \sum_{x_0,x_1} \mu(x_0)\,\nu(x_1)\,p^{{\rm base},\,|x_0,x_1}, \qquad p^\star_0 = \mu, \quad p^\star_1 = \nu \ \text{ by construction}, \tag{48}$$

whose jump intensity from x to y, given the endpoint, is the h-transform $\big(p^{\rm base}_{1|t}(x_1 \mid y)/p^{\rm base}_{1|t}(x_1 \mid x)\big)b_t(y, x)$. This is Lecture 2's reciprocal class on a chain: the two marginals are right by construction, but the intensity reads x_1, so $p^\star$ is not a Markov chain.

Theorem 5.2 (The Markovian projection keeps every marginal)**.** *Let the rate be the conditional mean of that intensity,*

$$u_t^\star(y,x) = \mathbb{E}_{p^\star}\Big[\frac{p_{1|t}^{\mathrm{base}}(X_1 \mid y)}{p_{1|t}^{\mathrm{base}}(X_1 \mid x)}\, b_t(y,x) \,\Big|\, X_t = x\Big]. \tag{49}$$

Then the chain with rate $u^\star$ started at $X_0 \sim \mu$ has law $p_t^\star$ at every t; in particular $X_1 \sim \nu$.

Proof. Let F be a test function. Applying theorem 1.4 under $p^\star$ — whose intensity is the pinned one — and then the tower property, which is legitimate because F reads only X_t,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}_{p^\star}\big[F(X_t)\big] &= \mathbb{E}_{p^\star}\Big[\sum_{y\neq X_t} \frac{p_{1|t}^{\mathrm{base}}(X_1 \mid y)}{p_{1|t}^{\mathrm{base}}(X_1 \mid X_t)}\, b_t(y,X_t)\,\big(F(y)-F(X_t)\big)\Big] \\ &= \mathbb{E}_{p^\star}\Big[\sum_{y\neq X_t} u_t^\star(y,X_t)\,\big(F(y)-F(X_t)\big)\Big].\end{aligned}$$

Since this holds for every F, the family $p_t^\star$ solves the master equation (2) of the rates $u^\star$; on a finite $\mathcal{X}$ that equation has a unique solution from a given initial condition, so the chain with rate $u^\star$ started at μ has exactly those marginals. This is why the method samples ν at all — everything below only fits $u^\star$. □

5.2 The Label

Theorem 5.3 (The rate is one conditional expectation)**.** *For the reciprocal measure (48) over $\mu \otimes \nu$ on $\mathcal{X} = \mathbb{Z}_N^D$,*

$$u_t^\star(y,x) = b_t(y,x)\,\mathbb{E}_{p^\star}\Big[e^{\,E(X_1)-E(X_1+y-x)} \cdot \frac{p_{1|0}^{\mathrm{base}}(X_1 \mid X_0)}{p_{1|0}^{\mathrm{base}}(X_1+y-x \mid X_0)} \,\Big|\, X_t = x\Big], \tag{50}$$

the first factor being the energy difference and the second the reference read end to end. Both are computed from the simulated triple (X_0, X_t, X_1), and Z cancels.

Proof. Write the flux out from (48), splitting each bridge at t:

$$p_t^\star(x)\,u_t^\star(y,x) = b_t(y,x)\sum_{x_0,x_1}\mu(x_0)\,\nu(x_1)\,\frac{p_{t|0}^{\mathrm{base}}(x \mid x_0)\,p_{1|t}^{\mathrm{base}}(x_1 \mid y)}{p_{1|0}^{\mathrm{base}}(x_1 \mid x_0)}.$$

Reindex the endpoint by the move, $x_1 \to x_1 + y - x$. By additivity (33) only differences matter, so $p_{1|t}^{\mathrm{base}}(x_1 + y - x \mid y) = p_{1|t}^{\mathrm{base}}(x_1 \mid x)$ and the y-dependence leaves the kernel; multiplying and dividing by $\nu(x_1)$ and by $p_{1|0}^{\mathrm{base}}(x_1 \mid x_0)$ restores the reciprocal measure inside the sum, and dividing by $p_t^\star(x)$ turns it into a conditional expectation:

$$u_t^\star(y,x) = b_t(y,x)\,\mathbb{E}_{p^\star}\Big[\frac{\nu(X_1+y-x)}{\nu(X_1)} \cdot \frac{p_{1|0}^{\mathrm{base}}(X_1 \mid X_0)}{p_{1|0}^{\mathrm{base}}(X_1+y-x \mid X_0)} \,\Big|\, X_t = x\Big].$$

Finally $\nu(X_1+y-x)/\nu(X_1) = e^{E(X_1)-E(X_1+y-x)}$, in which Z cancels. □

5.3 The Objective and the Loop

The matching loss Write $y = x^{d\leftarrow n}$ for the move being rated. With the draws

$$t \sim \mathrm{Unif}[0,1], \quad x_1 \sim \text{the current chain at } t=1, \quad x_0 \sim \mu, \quad x \sim p_{t|0,1}^{\mathrm{base}}(\cdot \mid x_0, x_1),$$

the label of theorem 5.3 is regressed exactly as in (45):

$$\min_{\Phi} \mathbb{E}_{t,\,x_0,\,x_1,\,x} \sum_{d,\,n\neq x^d} \mathcal{D}\left(e^{\,E(x_1)-E(x_1+y-x)}\, \frac{p^{\text{base}}_{1|0}(x_1 \mid x_0)}{p^{\text{base}}_{1|0}(x_1+y-x \mid x_0)} \;\Big\|\; \Phi_t(x)_{d,n} \right). \tag{51}$$

The label stands first in $\mathcal{D}$, so by the Bregman property the minimizer is its conditional mean — which is $u^\star/b_t$ by (50), that is $u = u^\star$.

Theorem 5.4 (The target is a fixed point)**.** *In training, x_1 comes from the chain being fitted, not from ν. Suppose nevertheless that the current rate sends μ to ν at $t=1$. Then the training draws are $x_0 \sim \mu$ and $x_1 \sim \nu$ independently, so their glued law is exactly $p^\star$ of (48); the regression (51) therefore returns $u^\star$, which sends μ to ν again by theorem 5.2.*

Proof. x_0 is drawn afresh, independently of the simulation, so the pair is a product measure whose second marginal is ν by hypothesis — which is the sampling in (48). The two cited results then apply verbatim. □

Remark (Consistency, not convergence)**.** As in the continuous case, this says only that ν reproduces itself. Whether the loop reaches it from a cold start, and whether it is the only fixed point, is open; damping the update is what makes it behave in practice.

The memoryless case If the reference has mixed by $t=1$, then $p^{\text{base}}_{1|0}(\cdot \mid x_0)$ is flat, the second factor of (50) is 1, and the label is the bare energy difference:

$$u^\star_t(y,x) = b_t(y,x)\, \mathbb{E}_{p^\star}\left[e^{\,E(X_1)-E(X_1+y-x)} \,\Big|\, X_t = x \right]. \tag{52}$$

The corrector $\widehat{\Phi}$ of algorithm 5 is then gone: *one network, one loss, nothing alternating.* At a finite rate the second factor is not 1, but it is explicit, so there is no reason to drop it. What is given up, in either case, is the bridge itself — only the two marginals are kept.

Algorithm 6 The discrete bridge matching sampler (Blessing et al., 2026b)

1: τ-leap K trajectories with the frozen rates $\mathrm{sg}[\,b_t\Phi\,]$ and keep the endpoints x_1.
2: One batched call gives $E(x_1)$ and the $D(N-1)$ energies of its single-site substitutions — the same budget as the soft label of (29).
3: Draw a fresh $x_0 \sim \mu$ and a time t, then bridge: $x \sim p^{\text{base}}_{t|0,1}(\cdot \mid x_0, x_1)$. The fresh x_0 is what makes the coupling $\mu \otimes \nu$: the model's own start is discarded.
4: Take a gradient step on (51), moving Φ only part of the way to the new fit, and return to step 1.

6 Extension to Diffusion Language Models

Two design questions On $\mathbb{Z}_N^D$ the adjoint identity (43) used $x_1 + y - x$ — arithmetic on token identifiers, which a vocabulary does not have. Two questions follow, and they are independent: (i) which symmetry replaces the shift, and (ii) which coordinates should the network predict? The group builds the target; the coordinates carry the controller.

Theorem 6.1 (The adjoint identity under any reference symmetry)**.** *Let a group G act on $\mathcal{X}$ with the reference G-equivariant, $p^{\text{base}}_{t|s}(gz \mid gx) = p^{\text{base}}_{t|s}(z \mid x)$ for all $g \in G$ and $s < t$, and let $g \in G$ satisfy $gx = y$. Then*

$$\frac{\varphi_t(y)}{\varphi_t(x)} = \mathbb{E}_{p^\star_{1|t}(x_1|x)}\left[\frac{\varphi_1(g\,x_1)}{\varphi_1(x_1)} \right]. \tag{53}$$

Proof. By the pull equation, substituting $z = gw$ and using $y = gx$ and equivariance,

$$\varphi_t(y) = \sum_z p^{\text{base}}_{1|t}(z \mid y)\varphi_1(z) = \sum_w p^{\text{base}}_{1|t}(gw \,|\, gx)\varphi_1(gw) = \sum_w p^{\text{base}}_{1|t}(w \mid x)\,\varphi_1(gw),$$

the substitution being a bijective change of variables. Dividing by $\varphi_t(x)$ and inserting the Doob form of the endpoint law, $p^\star_{1|t}(w \mid x) = p^{\text{base}}_{1|t}(w \mid x)\varphi_1(w)/\varphi_t(x)$, gives (53). The cyclic shift $g : x_1 \mapsto x_1 + y - x$ of theorem 4.5 is one instance. $\square$

The permutation symmetry of a vocabulary Take the reference to be the uniform chain (32) of Section 4, with the noise schedule γ_t introduced there. Each coordinate then moves independently under the generator $\gamma_t(U - I)$, where U is the uniform stochastic matrix on $[N]$; since $U^2 = U$, exponentiating gives

$$p^{\text{base}}_{1|t}(m \mid n) = \rho_t\, \mathbf{1}\{m = n\} + \frac{1 - \rho_t}{N}, \qquad \rho_t := e^{-\int_t^1 \gamma_s\, \mathrm{d}s} \;\in\; (0, 1]. \tag{54}$$

So the reference distinguishes only "same token" from "different token". It is therefore equivariant under $G = S_N^D$, relabelling each coordinate's vocabulary by $(g_\sigma x)^d = \sigma_d(x^d)$, and theorem 6.1 applies with g the transposition $\tau^{(d)}_{x^d,n}$ that swaps the two values $m = x^d$ and n at coordinate d and does nothing else:

$$\frac{\varphi_t\big(x^{d\leftarrow n}\big)}{\varphi_t(x)} = \mathbb{E}\left[\frac{\varphi_1\big(\tau^{(d)}_{x^d,n} X_1\big)}{\varphi_1(X_1)} \;\middle|\; X_t = x\right]. \tag{55}$$

The ratio from the logits For the controller, a language model already emits a per-coordinate row of logits $f^\theta_t(x)_{d,\cdot}$, and (54) converts it into the ratio the rate needs. Writing $\varphi_t(x) = \sum_{x_1} p^{\text{base}}_{1|t}(x_1 \mid x)\varphi_1(x_1)$ and factorizing the kernel at $x^{d\leftarrow n}$,

$$p^{\text{base}}_{1|t}\big(x_1 \,\big|\, x^{d\leftarrow n}\big) = \Big[\rho_t \mathbf{1}\{x_1^d = n\} + \tfrac{1-\rho_t}{N}\Big] \prod_{j \neq d} p^{\text{base}}_{1|t}\big(x_1^j \,\big|\, x^j\big),$$

so that if the row is trained to satisfy $\operatorname{softmax}(f^\theta_t(x)_{d,\cdot})_n \propto \sum_{x_1 : x_1^d = n} \varphi_1(x_1) \prod_{j\neq d} p^{\text{base}}_{1|t}(x_1^j \mid x^j)$, then

$$\frac{\varphi_t\big(x^{d\leftarrow n}\big)}{\varphi_t(x)} = \frac{\rho_t \operatorname{softmax}\big(f^\theta_t(x)_{d,\cdot}\big)_n + (1-\rho_t)/N}{\rho_t \operatorname{softmax}\big(f^\theta_t(x)_{d,\cdot}\big)_{x^d} + (1-\rho_t)/N}. \tag{56}$$

Remark (Which distribution the row belongs to)**.** The logits in (56) are attached to φ_1, not to the target: by (37), $\pi_{\varphi_1} \propto \varphi_1 = \nu/\widehat{\varphi}_1$, so only when $\widehat{\varphi}_1$ is constant — the memoryless case — is the row ν's own. The group action builds the adjoint-matching target and the logits carry the controller; the two are independent and can be used together. What is *not* removed by either is the dynamical stiffness of the bridge, or the variance of the adjoint target itself.

Summary

	learns	**the reward enters as**	**needs**
Discrete Feynman–Kac (§2)	nothing	$e^{\beta_t[r(y)-r(x)]}$	a pretrained model
Weighted TCSM (§3)	the rate u^θ	$\beta r(x_1)$, a weight	a pretrained model
Proximal cross entropy (§3)	the rate u^θ	a tempered weight	a pretrained model
Soft cross entropy (§3)	the row C^θ	an exact local label	an energy, a mask
Discrete ASBS (§4)	Φ and $\widehat{\Phi}$	$e^{E(x_1)-E(z)}$	an energy, a group
Discrete bridge matching (§5)	Φ alone	$e^{E(x_1)-E(x_1+y-x)}$	an energy, a group

Table 1 Five ways to sample $p_1^\star \propto p_1^{\text{base}} e^{\beta r}$, or $\nu \propto e^{-E}$, on a finite state space. The first three carry the same object — the Radon–Nikodym derivative of the target over what is simulated, theorem 2.1 — as a weight, as an importance weight for training, or tempered. The last three replace it by a label that is exact pointwise.

$$\text{One tilt and one tilted rate run through the lecture:} \qquad U_t(y,x) = e^{\,r_t(y)-r_t(x)}\, b_t(y,x),$$

corrected at test time (§2), regressed from weighted endpoints (§3), or solved for as a bridge (§4–§5).

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